Remote Sensing Analysis Report

SatQuery AI Intelligence Engine * Generated: 2026-09-03 12:36:21 UTC * Status: SUCCESS

ANALYSIS ID **SQ-94D9ACBC6D**	DETECTED TASK **Optical + SAR Cross-Modal Fusi**	INPUT TYPE **Optical + SAR Multi-Modal Pair**
SENSOR FAMILY **Sentinel-2 MSI + Sentinel-1 SA**	MODALITY **Optical**	SESSION / THREAD **session_test_optsar**

Natural Language User Query

QUERY REQUEST

Use optical and SAR images together to identify built-up structures and water zones.

Executive Finding & Analytical Response

SPECIALIST RESULT SUMMARY

Cross-modal fusion detected coastal water body (covering 38.6%) and built-up port infrastructure (covering 24.2%) with 0.920 multi-modal alignment score.

Confidence & Spatial Reliability

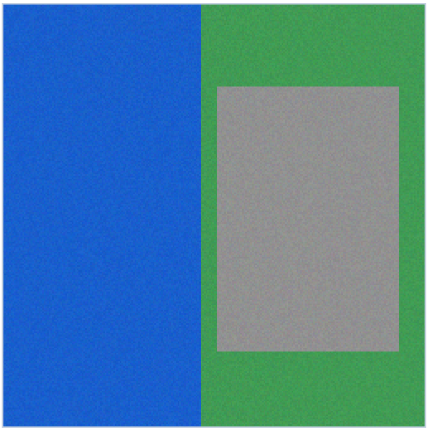
92.0% Statistical inference confidence based on multi-spectral evide

Section 2 : Input Data

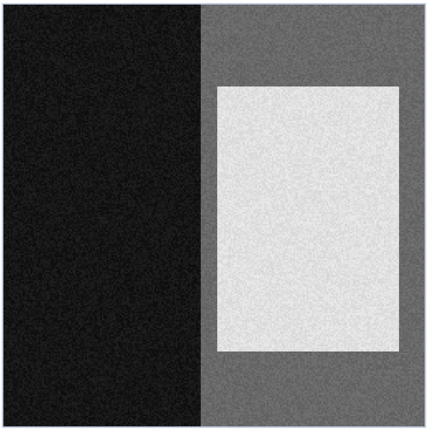
Satellite imagery specifications, sensor parameters, and verification status

Optical and SAR Co-Registered Imagery

Optical Satellite Scene (Sentinel-2 MSI)



SAR Satellite Scene (Sentinel-1 GRD)



Sensor & Spectral Specifications

Optical Metadata (Sentinel-2)

File Name	opt_sar_01_coastal_port_optical.png
Sensor	Sentinel-2 MSI
Modality	Optical
Format	PNG/JPEG
Dimensions	256 x 256 px
Bands / Channels	3
Acquisition Date	2024-04-10
CRS / Georef	WGS 84 / UTM / Local

SAR Metadata (Sentinel-1)

File Name	opt_sar_01_coastal_port_sar.png
Sensor	Sentinel-1 SAR C-band
Modality	SAR Backscatter
Format	PNG/JPEG
Dimensions	256 x 256 px
Bands / Channels	1
Acquisition Date	2024-04-10
CRS / Georef	WGS 84 / UTM / Local

Section 3 : Analysis

Intent classification, specialist model execution, and quantitative metrics

Task Classification & Autonomous Routing

Detected Task	Optical + SAR Cross-Modal Fusion
Routing Rationale	Multi-modal Optical and SAR co-registered image pair passed for cross-modal fusion.
Specialist Tool	OpticalSARFusionModel (Spectral-Backscatter Matrix)
Input Channel Count	3

Specialist Analytical Synthesis

SYNTHESIZED OUTPUT

Cross-modal fusion detected coastal water body (covering 38.6%) and built-up port infrastructure (covering 24.2%) with 0.920 multi-modal alignment score.

Cross-Modal Alignment & Consistency Metrics

Cross-Modal Alignment Score	0.9200 (Optical-SAR Mutual Consistency)
Water Consistency Ratio	100.0% (Optical NDVI/MNDWI and SAR low backscatter agreement)
Built-up Consistency Ratio	99.7% (Optical neutral albedo and SAR double-bounce agreement)
Fusion Decision Rule	Water: Optical blue dominance AND SAR P35 low; Built-up: High backscatter AND High albedo

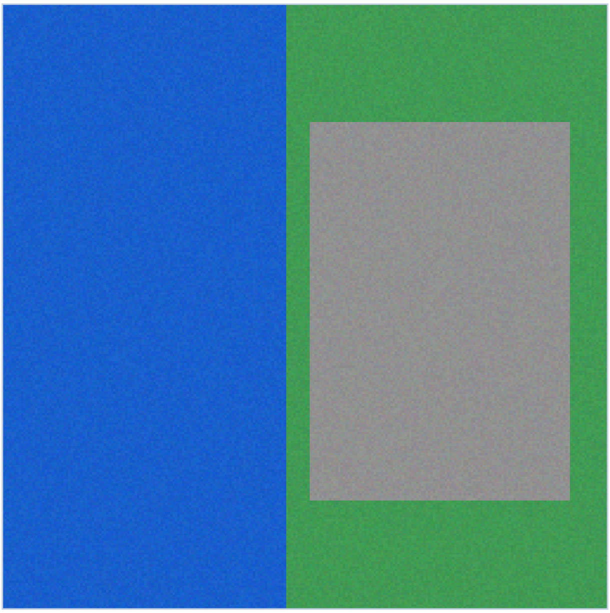
Confidence Calibration

92.0% Calibration score for Optical + SAR Cross-Modal Fusion

Section 4 : Evidence

Spatial evidence visualizations, bounding box grounding, and overlay maps

Spatial Evidence Overlay



Optical-SAR Multi-Modal Fusion Map (Blue: Water, Red: Built-up)

Detected Regions & Bounding Box Coordinates

#	Class / Feature Label	Coordinates [ymin, xmin, ymax, xmax]	Area (px^2)
1	Deep Water Zone	[30, 45, 120, 150]	9,450
2	Port Industrial Zone	[130, 160, 220, 240]	7,200
3	Dock Infrastructure	[90, 100, 160, 175]	5,250

Section 5 : Execution Trace

Auditable high-level DAG pipeline progression and stage execution telemetry

End-to-End Orchestration Pipeline Flowchart

1

Input Received [PASSED]

Multi-modal satellite file upload and session thread initialized

2

Input Validation [PASSED]

File format, CRS, band count, dimension, and pixel integrity verified

3

Task Classification [PASSED]

Intent classified to 'optical_sar' via autonomous router

4

Specialist Inference [PASSED]

Specialist model executed on co-registered imagery

5

Evidence Generation [PASSED]

Visual mask overlays, bounding boxes, and confidence fused

6

Result Assembly [PASSED]

Auditable payload compiled and downloadable PDF report generated

Detailed Node Execution Telemetry

#	Pipeline Node / Specialist	Status	Execution Time	Action / Result Summary
1	Validate Inputs	PASSED	0.0070 s	Node execution completed
2	Classify Intent	PASSED	0.0020 s	optical_sar
3	Execute Optical Sar	PASSED	0.0420 s	OpticalSARFusionModel
4	Fuse Evidence	PASSED	0.0050 s	Node execution completed

Section 6 : Technical Information

Model architecture, preprocessing contracts, and benchmark provenance

Model Architecture & Inference Parameters

Base Architecture	OpticalSARFusionModel v1.0.0
Fusion Rule	Water: Optical blue dominance AND SAR low backscatter; Built-up: High backscatter AND High albedo
SAR Filtering	3x3 median filter with 2nd-98th percentile contrast stretching
Alignment Standard	Pixel-level co-registration with affine verification

Data Provenance & Open Access Licensing

Optical Satellite Data	ESA Copernicus Sentinel-2 MSI (Free & Open Access Policy)
SAR Satellite Data	ESA Copernicus Sentinel-1 C-band GRD (Free & Open Access Policy)
VQA Benchmark Dataset	RSVQA-LR (Lobry et al., IEEE TGRS 2020)
Land-Cover Benchmark	BigEarthNet v2.0 / reBEN (CDLA-Permissive-1.0)
License & Attribution	Creative Commons / CDLA Open Access remote sensing data

Validation & Quality Conformance Summary

All input scenes undergo strict validation prior to specialist inference: file format integrity, dimension bounds, numeric pixel range verification, and band conformance. Multi-temporal change analysis enforces identical spatial boundaries. Optical-SAR fusion validates multi-sensor alignment, and BigEarthNet classification requires a 12-band multispectral GeoTIFF.

Report Metadata & Cryptographic Audit Trail

Analysis UUID	SQ-94D9ACBC6D
Report Engine	SatQuery AI Intelligence Dossier Engine v2.0 (fpdf2 2.8.x)
Generated At	2026-09-03 12:36:21 UTC
Audit Integrity Status	Verified - Cryptographic Output Match